OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California



Welcome to the **Future Technologies Symposium**

Dr Jon Summers, Chair of FTS

Scientific Lead in Data Centers at Research Institutes of Sweden

Adjunct Professor in Fluid Mechanics at Luleå Technical University, Sweden

Visiting Professor in Thermofluids at University of Leeds, UK



OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

Future Technologies Symposium Schedule

08:00 – 08:20	FTS: **Chair Keynote**
08:20 – 08:30	FTS: **Sponsor presentation**
08:30 – 08:50	FTS: **Awards Ceremony**
08:50 – 09:25	FTS: **AI / HPC** 2 presentations
09:30 – 09:45	FTS: **POSTER SESSION BREAK**
09:45 – 10:15	FTS: **AI / HPC** 5 Lightning Talks
10:20 – 10:55	FTS: **Data Center Sustainability** 2 presentations
11:00 – 11:30	FTS: **Data Center Sustainability** 5 Lightning Talks
11:30 – 12:30	FTS: **Lunch Break**
12:30 – 17:00	FTS: **Quantum**

Thanks to the FTS Global 2026 Sponsors

SAMSUNG

cerabyte

The Age of Physical Limits: Can AI overcome the laws of nature, or merely help humankind navigate them?



Dr Jon Summers, Chair of FTS

Scientific Lead in Data Centers at Research Institutes of Sweden

OCP FUTURE TECHNOLOGIES SYMPOSIUM

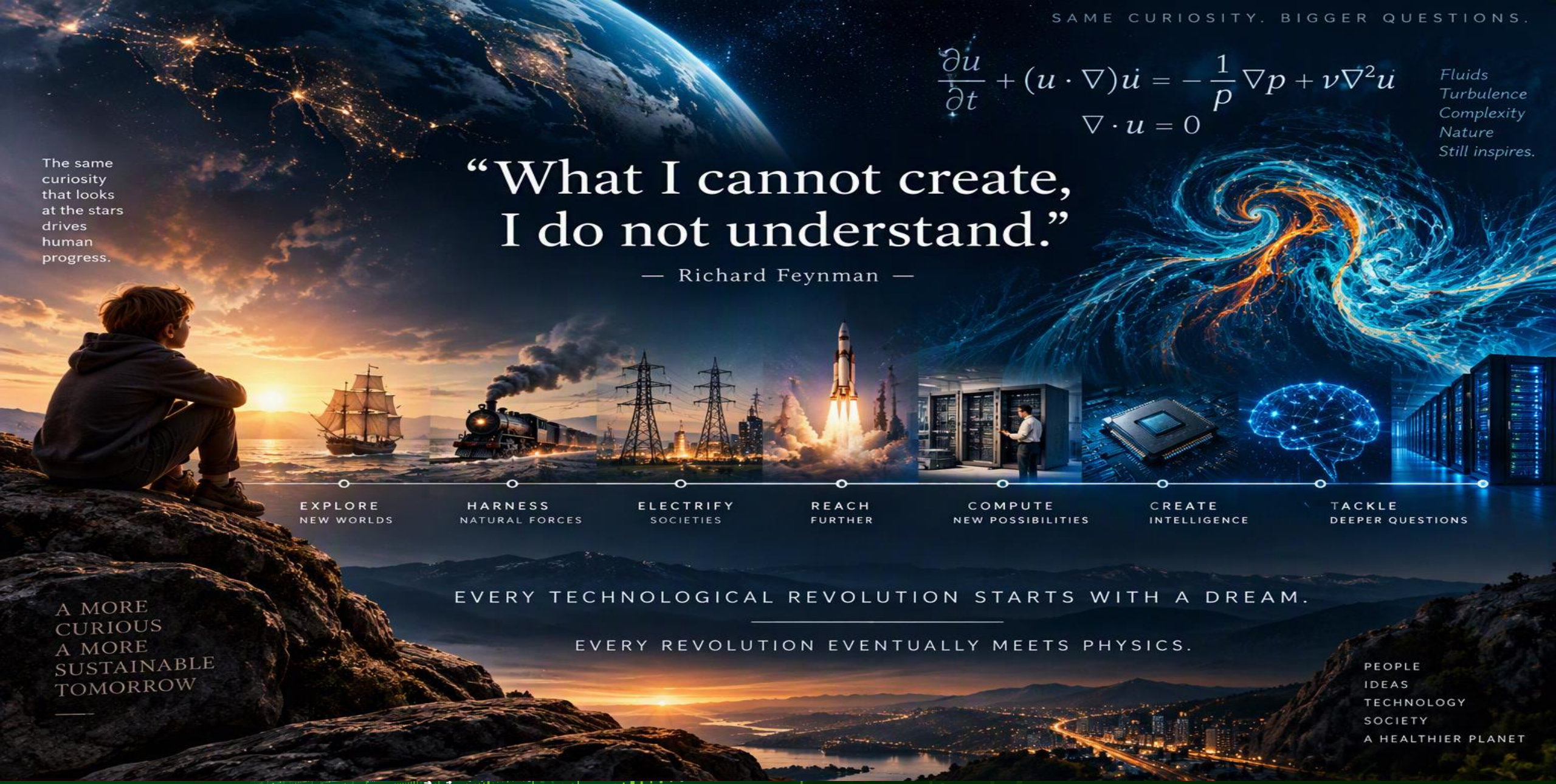
October 14, 2026
San Jose, California

RI.SE Research Institutes of Sweden



OCP FUTURE TECHNOLOGIES SYMPOSIUM | 2026

Scaling Innovation for the AI Era.



OCP FUTURE TECHNOLOGIES SYMPOSIUM | 2026

Scaling Innovation for the AI Era.

Every Revolution Eventually Encounters Physics!

- The **scaling wall**: pure lithography-enabled scaling is slowing down, and quantum effects begin interfering with the operation of microchips.
- The **memory wall**: system performance is confronted with data path limitations between the cores and the memory.
- The **power wall**: it is becoming more challenging to bring power into the chip and extract heat from the chip package.
- The **sustainability wall**: the manufacturing of semiconductor devices contributes to an increasing environmental footprint.
- The **cost wall**: chip manufacturing costs may explode with the complexity increase, together with the costs for design and process development.

Source: https://www.imec-int.com/en/articles/smaller-better-faster-imec-presents-chip-scaling-roadmap

The Infrastructure Beneath Intelligence.

01
AI Models
LLMs & Neural Nets

02
Silicon Chips
GPUs & Accelerators

03
Servers
High-Density Racks

04
Data Centers
Hyperscale Plants

05
Power Grid
HV Transmission

06
Energy System
Nuclear / Gas / Hydro

System Bottleneck Shift: Value constraints have migrated downwards from algorithmic models to high-voltage grid interconnection capacity.

Electricity
Gigawatt-scale continuous baseload power.

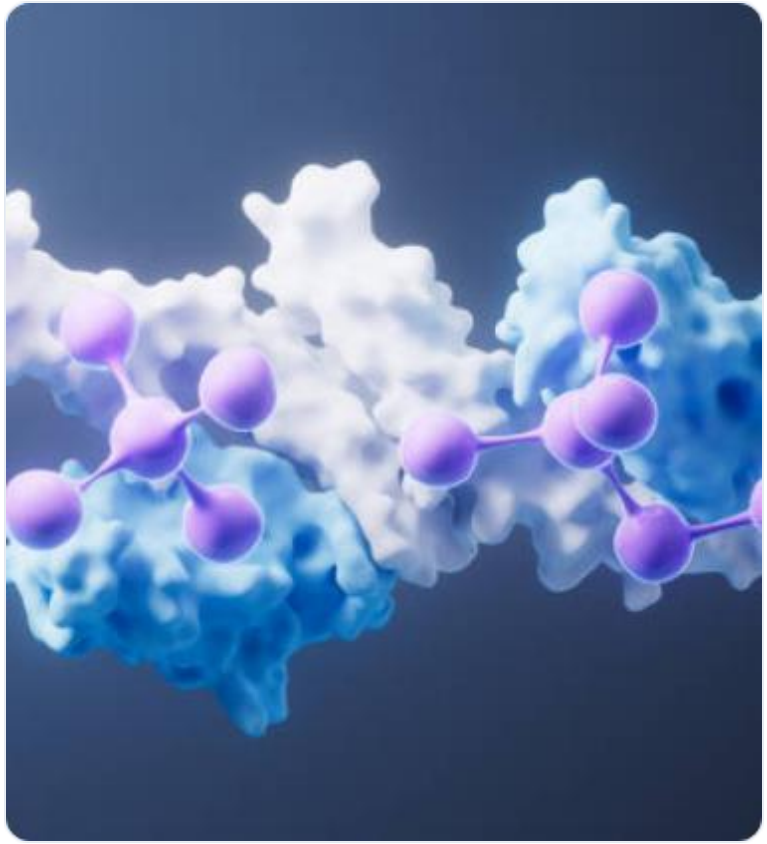
Cooling
Direct-to-chip & liquid immersion loops.

Materials
Pure silicon, copper, & rare earth elements.

Water
Millions of gallons for heat exchange.

Infrastructure
Fiber backbones & substation real estate.

Extraordinary Breakthroughs Across Domains.



Source: www.istockphoto.com

AlphaFold & Biology

Predicting 200M+ protein structures with atomic precision.

Materials Discovery

Accelerating novel crystal and battery compound synthesis.

Software Generation

Automating code synthesis and complex engineering logic.

Literature Analysis

Synthesizing millions of scientific papers into insight.

Navier–Stokes Equations

$$\underbrace{\frac{\partial \mathbf{u}}{\partial t}}_{\text{Unsteady acceleration}} + \underbrace{(\mathbf{u}\cdot\nabla)\mathbf{u}}_{\text{Nonlinear advection}} = \underbrace{-\frac{1}{\rho}\nabla p}_{\text{Pressure gradient}} + \underbrace{\nu\nabla^2\mathbf{u}}_{\text{Viscous diffusion}} + \underbrace{f}_{\text{External forces}}$$

Existence and smoothness?

Global regularity in 3D
Finite energy
No singularities?
Unique solution?
For all time?

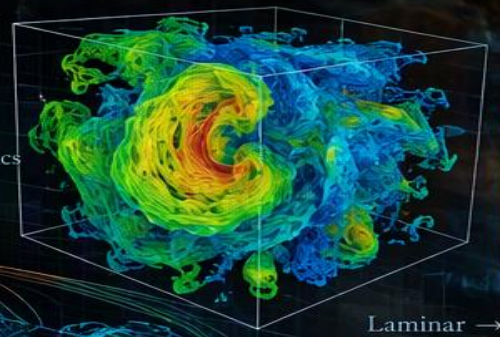
CLAY MATHEMATICS INSTITUTE

MILLENNIUM PROBLEM

ONE OF THE SEVEN GREATEST UNSOLVED PROBLEMS IN MATHEMATICS

Turbulence

Vortices
Instability
Energy cascade
Multiscale dynamics
Chaos



Laminar → Turbulent
Transition
Boundary layers
Separation
Vortex shedding
Three-dimensional flow



Can AI solve it?

Predict?
Understand?
Prove?

Atmosphere
Oceans
Aerospace
Energy systems
Blood flow
Cooling technologies
Data centers
Our physical world

From equations … to the motion of our world



OCP FUTURE TECHNOLOGIES SYMPOSIUM | 2026

Scaling Innovation for the AI Era.

Has AI Solved Navier-Stokes?

Central Epistemological Dilemma: **Is prediction the same as understanding?**

Modified vs. True Equations

Claims rely on simplified Euler systems or unphysical forcing functions—not the 3D Millennium existence statement.

Formalization Gaps (Lean)

Automated translation into Lean formal proof code altered the underlying mathematical statement being verified.

Emulation ≠ Analytical Proof

Numerical simulations and curve fitting observe turbulence but cannot establish continuous smooth existence.

PR vs. Domain Peer Review

Corporate benchmarking press releases frame unsolved math problems as PR milestones without PDE rigor.

Key Takeaway: **Statistical curve-fitting mimics fluid behaviour but cannot substitute for analytical mathematical rigour.**

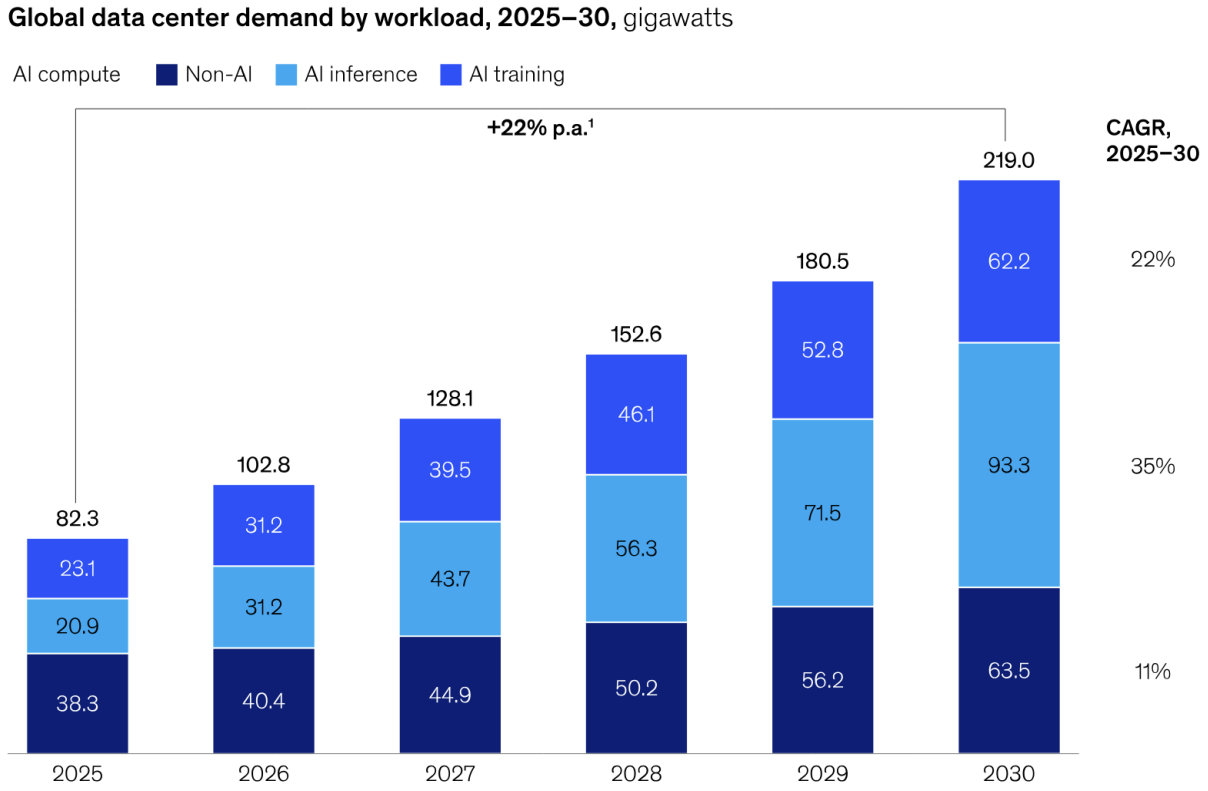
Navier–Stokes Millennium Problem Remains Officially Unsolved!

Meanwhile Reality Sends The Bill

While we debate AI's intellectual power:

- energy demand rises
- cooling demand rises
- infrastructure demand rises

Intelligence has a physical cost.



Source: McKinsey 2025

The New Grand Challenge

Not

Can we build smarter AI?

But

Can we build intelligent systems that remain compatible with energy, water materials and climate constraints?

Emerging technologies:

- Photonic computing
- Quantum computing
- Reversible computing
- Superconducting systems
- Advanced cooling
- AI-designed materials

All responding to physical limits.

Final Thoughts

- We live in an extraordinary time.
- We may soon create machines capable of discovering solutions to problems that have challenged humanity for centuries.
- Yet many of those discoveries will concern a deeper truth:
  - The future is not constrained by our imagination.
  - It is constrained by energy, matter and the laws of nature.

- The future of computing is no longer a computing problem.
- It is an energy, infrastructure and sustainability problem.

OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

